

■ CortexKPI — Autonomous Executive Storytelling Engine

Accenture Innovation Challenge 2026 | Problem Statement 3: BusinessIntelligence.ai

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1. Executive Overview & Problem Statement

When high-stakes enterprise metrics drop (e.g. an unexpected **\$190,000/day revenue loss** in APAC), executive leadership and engineering teams spend 6+ hours sifting through fragmented Datadog APM charts, Jira pull requests, Zendesk support tickets, and Slack alerts trying to answer: *What happened, why it happened, and how to fix it?*

CortexKPI is an autonomous AI diagnostic engine that unifies seasonal Bayesian anomaly detection, causal dependency graph decomposition, multimodal RAG log vectorization, and epistemic safeguards into an interactive executive dashboard with **1-Click automated mitigation**.

2. Architectural Pillars (The 4 ML Layers)

Layer / Module	Underlying ML Algorithm	Core Purpose & Enterprise Value
■ Step 1: Metric Timeline & Anomaly Detection	Seasonal STL Decomposition + Bayesian Baselineing (Student-t) + Isolation Forest Unsupervised ML	Eliminates false alarms from normal weekend dips; calculates dynamic S
■ Step 2: Causal Metric Dependency Chain	Structural Causal DAG (NetworkX) + Mathematical Variance Attribution + Counterfactual Intervention Matrix	Decomposes Revenue into Sessions × Conversion × AOV. Isolates root
■ Step 3: Multimodal RAG Evidence Fusion	TF-IDF N-Gram Vectorizer + Cosine Similarity + Exponential Temporal Decay	Scans unstructured Jira deployments, Zendesk support tickets, and Slack
■ Step 4: Executive Verdict & 1-Click Fix	4-Pillar Epistemic Safeguards + Dual-Engine Synthesis + Local Qwen 3.8 / LLM AI Hook	Generates concise 3-bullet executive briefings; categorizes facts vs gaps

3. Dual-Mode UX & Persona-Aware Dashboard

- **Simple View vs Technical Diagnostics:** Toggle between plain-English executive summaries for leadership and deep statistical Z-score/distribution matrices for data scientists.
- ■ **C-Suite Persona:** Emphasizes daily financial impact (-\$190,107/day) and strategic ROI.
- ■■ **DevOps Persona:** Displays microservice commit hashes, pull requests, APM logs, and 1-click rollback buttons.
- ■ **BI Analyst Persona:** Displays variance attribution %, p-values, and cosine similarity match scores.
- **Interactive Node Inspector:** Clicking any metric in the causal tree immediately filters all system logs to that specific sub-system.

4. Real-World Datasets Ingested (17,000+ Total Records)

- **metrics_timeseries.csv:** 9,125 rows across 5 global regions (APAC, EU, NORTH_AMERICA, LATAM, EMEA) over 365 days.
- **jira_deployments.csv:** Live-crawled deployment logs across major cloud repositories (Stripe, React, FastAPI, Next.js).
- **zendesk_tickets.csv:** Customer care complaints with sentiment scores (-0.92 to +0.45).
- **slack_alerts.csv:** Datadog APM, PagerDuty, and Cloudflare DNS war room monitoring alerts.
- ■ **Custom CSV Portal:** Ingest any custom enterprise dataset dynamically via the top-header portal in <2 seconds.

5. Local Quickstart & Verification Suite

```
# 1. Clone & install
git clone https://github.com/abhishek9680/CortexKPI.git
cd CortexKPI
pip install -r requirements.txt

# 2. Launch FastAPI Server
python main.py # Dashboard live at http://127.0.0.1:8000

# 3. Run Verification Test Suite
python test_production_api.py
python test_custom_inputs.py
```